MATH 243: Discussion Worksheet 13

Topics: 16.4 Green's Theorem; 16.5 Curl and Divergence; 16.6 Parametric Surfaces and Their Areas

Please put away all electronic devices, including cell phones and calculators.

1. Use Green's Theorem to evaluate the line integral.
   (a) $\oint_C (y + e^{\sqrt[3]{x^5}})\,dx + (2x + \cos(y^2))\,dy$, where C is the positively oriented boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.
   (b) $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = (3 + e^{x^2})\,\mathbf{i} + (\arctan y + 3x^2)\,\mathbf{j}$ and C is the positively oriented boundary of the quarter annulus $1 \le x^2 + y^2 \le 4$ in the first quadrant.

2. Consider the force field $\mathbf{F}(x, y) = xy^2\,\mathbf{i} + 5x^2y\,\mathbf{j}$.
   (a) Determine whether $\mathbf{F}$ is conservative.
   (b) Find the work done by $\mathbf{F}$ in moving a particle around the positively oriented triangular path C with vertices $(0, 0)$, $(1, 1)$, and $(1, 2)$.

3. Compute the curl and divergence of each vector field.
   (a) $\mathbf{F}(x, y, z) = x^3yz^2\mathbf{j} + y^4z^3\mathbf{k}$
   (b) $\mathbf{F}(x, y, z) = \ln(2y + 3z)\mathbf{i} + \ln(x + 3z)\mathbf{j} + \ln(x + 2y)\mathbf{k}$
   (c) $\mathbf{F}(x, y, z) = (x^2 \ln(y + 1))\mathbf{i} + (y^2z^3)\mathbf{j} + \left(\dfrac{z}{y}\right)\mathbf{k}$

4. Find a parametric representation for the surface.
   (a) The part of the hyperboloid $4x^2 - 4y^2 - z^2 = 4$ that lies in front of the yz-plane.
   (b) The part of the cylinder $x^2 + z^2 = 9$ that lies above the xy-plane and between the planes $y = -4$ and $y = 4$.

5. Find an equation of the tangent plane to the surface at the given point.
   (a) The surface given by $x = u^2 + 1$, $y = v^3 + 1$, $z = u + v$ at the point $P(5, 2, 3)$.
   (b) The surface described by $\mathbf{r}(u, v) = u^2\,\mathbf{i} + 2u \sin v\,\mathbf{j} + u \cos v\,\mathbf{k}$ at the point corresponding to $(u, v) = (1, 0)$.

6. Find the area of the surface S that is the part of the paraboloid $y = x^2 + z^2$ that lies within the cylinder $x^2 + z^2 = 16$.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Green's Theorem. Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane, and let D be the region bounded by C. Assume that P and Q have continuous first-order partial derivatives on an open region containing D. Then

$$\oint_C P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA.$$

Divergence and Curl. Let

$$\mathbf{F}(x,y,z) = P(x,y,z)\,\mathbf{i} + Q(x,y,z)\,\mathbf{j} + R(x,y,z)\,\mathbf{k}.$$

Assume that all first-order partial derivatives of P, Q, and R exist. The **divergence** of $\mathbf{F}$ is

$$\operatorname{div}\mathbf{F} = \nabla\cdot\mathbf{F} = P_x + Q_y + R_z.$$

The **curl** of $\mathbf{F}$ is

$$\operatorname{curl}\mathbf{F} = \nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = (R_y - Q_z)\,\mathbf{i} + (P_z - R_x)\,\mathbf{j} + (Q_x - P_y)\,\mathbf{k}.$$

Tangent Planes to Parametric Surfaces. Suppose that S is given parametrically by

$$\mathbf{r}(u,v) = x(u,v)\,\mathbf{i} + y(u,v)\,\mathbf{j} + z(u,v)\,\mathbf{k}.$$

Let P_0 be a point on S with position vector $\mathbf{r}(u_0,v_0) = \langle x_0, y_0, z_0\rangle$. Then

$$\mathbf{n} = \mathbf{r}_u(u_0,v_0)\times\mathbf{r}_v(u_0,v_0)$$

is a normal vector to the tangent plane of S at P_0, provided this vector is nonzero.

Surface Area. Suppose that a smooth parametric surface S is given by

$$\mathbf{r}(u,v) = x(u,v)\,\mathbf{i} + y(u,v)\,\mathbf{j} + z(u,v)\,\mathbf{k},$$

where S is covered exactly once as (u,v) ranges over the parameter domain D. The surface area of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v|\; dA.$$

For the special case where S is given explicitly by $z = f(x,y)$, where $(x,y)\in D$ and f has continuous first partial derivatives, the surface area is

$$A(S) = \iint_D \sqrt{1 + f_x^2 + f_y^2}\, dA,$$

where D is the projection of S onto the xy-plane.

Suggested Textbook Problems
Section 16.4: 1-18, 21-23
Section 16.5: 1-12, 15-20
Section 16.6: 1-26, 33-50